

THE UNIVERSITY OF CHICAGO
DATA SCIENCE INSTITUTE

Configuring a Coding Agent

Outline

- What are coding agents?
- Connecting to an inference service
- Configuring a coding agent
    - Claude code
    - Pi

Before You Start (Prerequisites)

- A terminal (macOS Terminal, Windows Linux Subsystem, Linux shell)
- uv installed

(Optional) Docker

What is a coding agent?

- An agent that writes and edits code for you
- Plans and completes multi-step tasks on its own
- Runs and tests its work, then fixes mistakes, iteratively
- You review and approve the result







Connecting to a model

ALCF Inference Service

```
$ uv tool install "alcf-ai==0.10.0"
$ alcf-ai auth login
```

Login using your ALCF account

```
$ export ALCF_AI_TOKEN="$(alcf-ai auth get-access-token)"
```

```
$ alcf-ai chat -c minerva -m inkling-bf16 "hello"
```

Claude Code configuration

```
$ curl -fsSL https://claude.ai/install.sh | bash
```

```
$ vim ~/.claude/settings.json
```

```
$ claude
```

Claude Code configuration




```
$ curl -fsSL https://claude.ai/install.sh | bash
```

```
vim ~/.claude/
```

```
$ claude
```

```
{
  "env": {
    "ANTHROPIC_BASE_URL":
"https://inference-api.alcf.anl.gov/resource_server/minerva/api",
    "ANTHROPIC_AUTH_TOKEN": "<your-api-key-here>",
    "ANTHROPIC_DEFAULT_OPUS_MODEL": "inkling-bf16",
    "ANTHROPIC_DEFAULT_SONNET_MODEL": "inkling-bf16",
    "ANTHROPIC_DEFAULT_HAIKU_MODEL": "inkling-bf16"
  }
}
```

Pi configuration



```
$ npm install -g --ignore-scripts @earendil-works/pi-coding-agent
```

```
$ vim ~/.pi/agent/models.json
```

```
$ pi
```

Pi configuration



```
$ npm
```

```
$ vim
```

```
$ pi
```

```
{
 ...
 "providers": {
     "alcf-minerva": {
        "baseUrl":
"https://inference-api.alcf.anl.gov/resource_server/minerva/api/v1",
        "api": "openai-completions",
        "apiKey": "!alcf-ai auth get-access-token",
        "compat": {
          "supportsDeveloperRole": false,
          "supportsReasoningEffort": false
         },
         "models": [{ "id": "inkling-bf16" }, {"id": "nemotron-3-ultra"}]
       }
   }
   ...
}
```

Further configuration instructions

- Argonne Inference Service
  - https://docs.alcf.anl.gov/services/inference-endpoint
- American Science Cloud Model Access Gateway
  - https://amsc-docs-d762d2.gitlab.io/model-access-gateway/
- Claude Code
  - https://code.claude.com/docs/en/overview
- Pi
  - https://pi.dev/docs/latest